

Q1

$$-5 - 8 = -13$$

Q2

47 and 61 are the two primes in the list. $47 + 61 = 108$

Q4

$$3.5 \times 250000 = 875000 \text{ cm} = 8.75 \text{ km}$$

Q12a

$$a \times 8^2 + b = 181, \text{ so } 64a + b = 181. \text{ Trying } a = 2 \text{ (both } > 1): b = 181 - 128 = 53$$

Q12b

$$x^2 = 64 \text{ has two roots, } 8 \text{ and } -8, \text{ so the other solution is } x = -8$$

Q14

$$y = 5x + 2, \text{ so } 5x = y - 2, x = (y - 2) / 5. f^{-1}(x) = (x - 2) / 5$$

Q19

$$m = k / (t+2)^2. 0.64 = k / 25 \text{ so } k = 16. \text{ When } t = 8: m = 16 / 100 = 0.16$$